

Self-Supervised and Contrastive Representation Learning on STL-10

DATA 266 | Ritika Mukesh Neema | SID4 6638

1. Setup

All parts use **ResNet-18 with no pretrained weights** on STL-10 (96x96 images; 5,000 labeled train, 100,000 unlabeled, 8,000 test). Step 0 parameters follow Assignment 1: SID4=6638, SEED=6638, SLICE=638, HP_ID=2, CLS_A=8, CLS_B=5; all random seeds are set from SEED. Parts A, B and C share the **same 500-image labeled subset** (10% of train, stratified: 50 per class). Runs were done on a Colab T4 GPU with mixed precision. Augmentations run on the GPU and use the four SimCLR augmentations: random resized crop, horizontal flip, colour jitter, random grayscale.

2. Methods

Part A - Supervised. ResNet-18 plus a linear layer trained end-to-end on the 500 labeled images for 15 epochs (Adam, lr 1e-3, batch 64, light crop/flip augmentation).

Part B - Rotation SSL. The encoder is trained on all 100,000 unlabeled images for 15 epochs to predict a random rotation (0, 90, 180, 270 degrees) with a 4-way linear head (Adam, lr 1e-3, batch 256). The head is then removed, the encoder is frozen, and a linear classifier is trained on the 500 labels for 20 epochs. Encoder weights are checked unchanged after the probe.

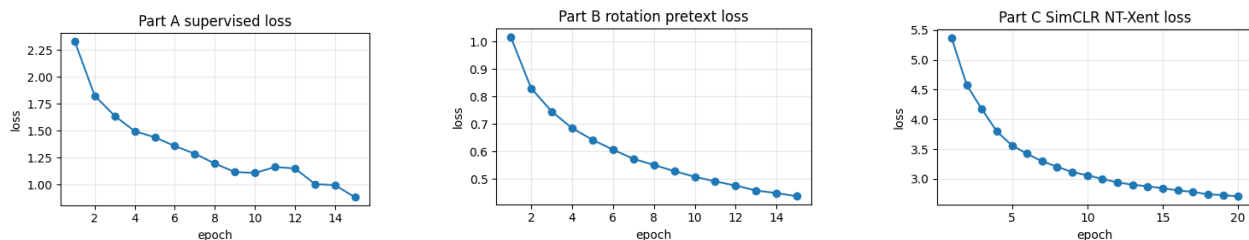
Part C - SimCLR. 20,000 unlabeled images, two augmented views per image, a two-layer MLP projection head (512-512-128), NT-Xent loss with cosine similarity and temperature $\tau = 0.2$, batch 256, 20 epochs. The projection head is removed, the encoder frozen, and a linear classifier trained on the 500 labels for 20 epochs (frozen-encoder check as in Part B).

Part D - Nearest neighbours. Test-set embeddings (512-d encoder output) are L2-normalised so a dot product is cosine similarity. Three query images from different classes (test indices 3808, 2548, 3442: cat, deer, bird) are used for all three encoders; the top-5 neighbours (query excluded) are shown with class labels. I also report kNN precision@5 over the entire test set.

3. Results

Model	Pretrain	Test accuracy	kNN precision@5
A Supervised, 500 labels	none	38.12%	36.36%
B Rotation SSL + frozen linear probe	15 ep, 100k imgs	49.31%	40.96%
C SimCLR ($\tau=0.2$) + frozen linear probe	20 ep, 20k imgs	53.80%	47.46%

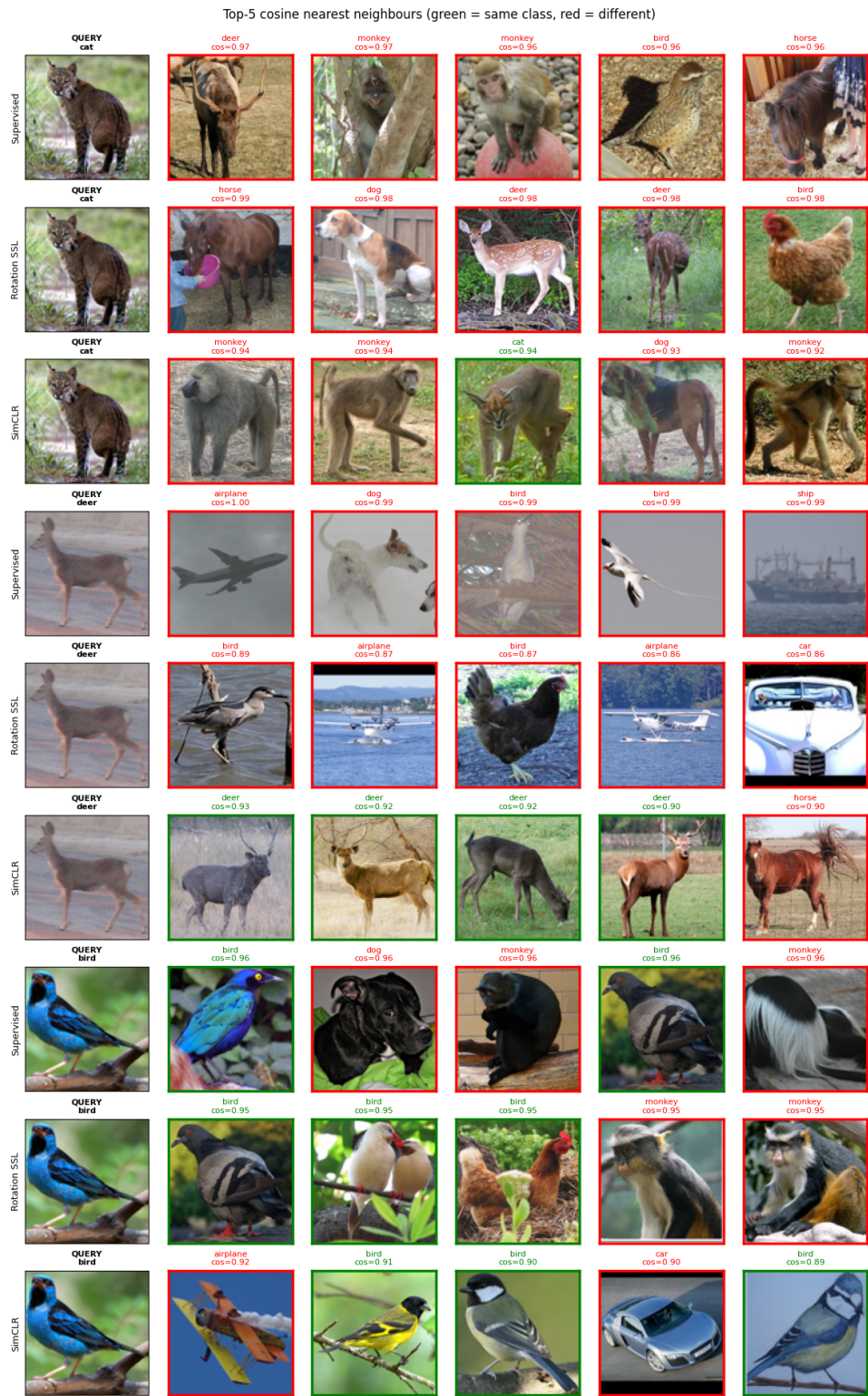
Rotation pretext accuracy reached 83.2% (chance 25%). SimCLR contrastive top-1 reached 80.4%. Test accuracy is on all 8,000 test images; results are from a single run (seed 6638).



Training curves: Part A supervised loss, Part B rotation loss, Part C NT-Xent loss.

4. Nearest-neighbour visualisation (Part D)

Rows are Supervised, Rotation SSL, SimCLR for each query. Green border = same class as the query, red = different class.



5. Discussion

Which model performed best, and why?

SimCLR (53.8%), then rotation (49.3%), then supervised (38.1%). SimCLR's objective asks for what a classifier needs: two heavily augmented views (crop, flip, colour jitter, grayscale) of one image must map close together while all other images are pushed away. Surviving those augmentations forces the encoder to ignore colour, position and scale and keep object-level content, so a linear layer on only 500 labels can separate the classes. SimCLR also used only 20k unlabeled images, 5 times fewer than rotation, and still won.

Why did the other models perform the way they did?

- **Supervised (38.1%):** 500 images at batch 64 is about 8 steps per epoch, so 15 epochs is only about 120 gradient updates from random initialisation. The loss was still falling (2.33 to 0.88), so this baseline is undertrained as well as data-starved. It never sees the 100k unlabeled images.
- **Rotation (49.3%):** the pretext task reached 83% accuracy, but it can be solved with cues that are not class-specific, such as sky and ground layout, horizon lines, background texture or black letterbox borders. The encoder learns scene geometry more than object identity, and final-layer features specialise to the pretext task, which limits transfer. It still clearly beat supervised, so the unlabeled data helped.
- **SimCLR (53.8%):** still not converged. The NT-Xent loss was still falling (2.70) and contrastive top-1 still rising (80%) at epoch 20, and 20k images with batch 256 (510 negatives per step) is small by SimCLR standards.

What changes could improve each model?

- **Supervised:** more update steps (batch 16-32 or more epochs), stronger augmentation, label smoothing or mixup, and a learning-rate schedule.
- **Rotation:** longer pretraining; add crop and colour augmentation so the task cannot use trivial cues; probe intermediate layers (layer3) instead of only the pooled output.
- **SimCLR:** train longer, use all 100k images, a larger batch or memory queue for more negatives, a cosine schedule with warmup (or LARS), tuned augmentations, and a sweep of tau around 0.1-0.5. An MLP probe or light fine-tuning on the 500 labels would also raise accuracy.

What did the nearest-neighbour visualisations reveal?

Across the three queries (15 neighbours per model), same-class neighbours were Supervised 2/15, Rotation 3/15 and SimCLR 8/15, matching the kNN precision@5 ordering.

- **Supervised** retrieves by low-level appearance: the grey deer query returns an airplane, a dog, two birds and a ship, all pale grey with low-contrast backgrounds. Its cosine similarities are all about 0.96-1.00, meaning the embeddings occupy a narrow cone and are barely discriminative.
- **Rotation** retrieves by scene and layout: the cat returns mostly animals in green, grassy scenes (deer, deer, chicken), and the deer returns birds and planes on water plus a car and plane with black letterbox bars, suggesting border and scene cues are used.
- **SimCLR** is the most semantic: 4 of the 5 neighbours of the deer are deer, with colour, background and pose varying. Its errors are plausible look-alikes (monkeys and a dog for the cat, a horse for the deer), though an airplane and a car also appear for the bird query.

Caveat: single run, one seed. With 500 labels, seed-to-seed variation could be a few points, so the ordering is clear but the exact gaps are not tight.